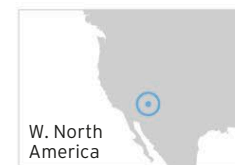


Meteor Crater

The best-preserved meteor impact site on Earth and evidence of an explosive 50,000-year-old collision whose effects can still be seen several miles away



A massive, bowl-shaped cavity carved into the Arizona Desert, Meteor Crater is one of the most studied meteorite craters on Earth. Its shape has been remarkably well preserved, and it is comparatively young for an impact site. Nearly 1 mile (1.6 km) across, more than 550 ft (168 m) deep, and with a circumference of 2½ miles (3.9 km), the depression known as Meteor Crater was initially believed to be an extinct volcano. It was not until the early 20th century that it was identified as having been created by an extraterrestrial impact, and details about its origins are still being discovered.

Around 50,000 years ago, a meteorite estimated at 130 ft (40 m) across and weighing 300,000 tons (272,000 tonnes) hurtled into Earth's atmosphere at 26,800 mph (43,120 kph)—10 times faster than gunshot. Half of it slammed intact into what is now the northern Arizona Desert, forming the crater; the rest fragmented before the collision, falling to the surface as a cloud of debris. Iron-rich meteorites weighing 1–1,000 lb (0.5–454 kg) have been found in a 6-mile (10-km) radius around the main impact site.

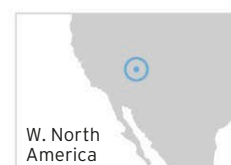
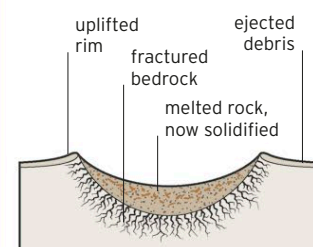


△ EXPLOSIVE IMPACT

Scientists estimate that the force released by the meteorite impact was greater than that generated by 20 million tons (18 million tonnes) of TNT.

CRATER FORMATION

The impact created when a single meteorite hits Earth often destroys the meteorite itself in a shock-wave-inducing explosion that fractures bedrock and melts surface rock. It leaves a bowl-shaped depression with a raised outer rim of ejected debris and a base lower than the surrounding landscape.



Mesa Arch

A dramatic natural archway formed on the edge of a high mesa

It looks like a bridge, but the 88-ft (27-m) long Mesa Arch, in southeastern Utah's Canyonlands National Park, is a pothole arch, created not by running water but by weathering and erosion. Water accumulated in a small depression in a sandstone mesa known as the Island in the Sky. Over time, the depression was enlarged by weathering and erosion into a larger pothole, undercutting more resistant rock at the cliff edge and eventually forming what is now thought to be the most-photographed natural arch in the world. Although pothole arches survive for thousands of years, they do eventually collapse. However, seismic research in 2015 indicated that Mesa Arch is stable—for now.

△ CLIFFTOP ARCH

Mesa Arch stretches across the top of the Island in the Sky mesa, a headland that rises 2,000 ft (600 m) above the base of Canyonlands National Park. Below it, the mesa wall drops vertically for more than 490 ft (150 m).